BUS 123 — MATH-M07-L01 — Sales, Excise & Property Taxes

BUS 123 · MATH-M07-L01 · SALES, EXCISE & PROPERTY TAXES

Course: Solving Business Problems with Technology · Fall 2026 **Track:** MATH · **Module:** M07 · **Lesson:** L01
Case Study Companies: Tidal Goods Co. · Meridian Advisory Group

1 · CONNECT TO PRIOR KNOWLEDGE

In the previous module, you learned how businesses set selling prices — applying trade discounts to reduce the retail price before quoting a buyer, and marking up costs to establish a profitable price point. You built those calculations in Excel using cell references so that any change in an input automatically updated the entire chain.

This lesson adds a critical layer: **government taxes.** Once a price is set, taxes are applied at specific points in a defined order — and getting that order wrong produces incorrect totals, incorrect revenue reporting, and potential legal liability. Today we examine three distinct tax types — sales, excise, and property — each of which places a different obligation on the business.

> 📘 **Training Jurisdiction Rules**
> Tax rules vary by location. To keep the lesson focused on business math, every example in this module uses one simplified training jurisdiction:
> - qualifying trade discounts reduce the sales-tax base;
> - shipping is added after sales tax;
> - when a scenario states that sales and excise taxes share a base, both use the original base cost;
> - property is taxed using the supplied assessment rate, and the property-tax rate is rounded to four decimal places before use.
>
> Treat these as the supplied rules for the lesson, not as universal tax advice.

2 · CORE CONCEPTS

Part 1 — Sales Tax

What Is Sales Tax?

Sales tax is a consumption tax imposed on specified retail sales. Rates and taxable items vary by jurisdiction. In this lesson, the problem statement always supplies the rate and the taxable-base rules you should use.

> ⚠️ **The Merchant's Role**
> When a business collects sales tax from a customer, those funds are **NOT revenue**. They are a **liability held in trust** for the state government. The business is acting as a collection agent. Misclassifying collected tax as earned revenue is a serious accounting error.

The Order of Operations

For this lesson's training jurisdiction, trade discounts reduce the taxable base and shipping is added after tax. Reversing the supplied sequence produces an incorrect result.

Step	Operation	Formula
1	Subtract trade discount from retail price	`Taxable = Price − Discount`
2	Apply tax rate to taxable subtotal	`Tax = Taxable × Rate`
3	Add shipping to get total price	`Total = Taxable + Tax + Shipping`

Worked Example — Tidal Goods Co.

Tidal Goods Co. sells a bulk supply order to a business customer for its own use. Retail price: $480.00. Trade discount: $80.00. Sales tax: 6.25%. Flat shipping: $15.00.

Item	Amount
Retail Price	$480.00
Less: Trade Discount	−$80.00
Taxable Subtotal	**$400.00**
× Sales Tax Rate (6.25%)	× 0.0625
Tax Owed	$25.00
Taxable Subtotal + Tax	$425.00
+ Shipping	+$15.00
Total Price	**$440.00**

Back-Calculating Actual Sales

When a retailer's register total includes both sales revenue and collected tax, the actual pre-tax revenue must be isolated:

```
Actual Sales = Total Receipts / (1 + Tax Rate)
```

Example: Tidal Goods Co.'s register shows $52,000 in total daily receipts at a 6.25% tax rate.

- Actual sales = $52,000 ÷ 1.0625 = **$48,941.18**
- The remaining **$3,058.82** belongs to the state.

Part 2 — Excise Tax

What Is Excise Tax?

Excise tax is a targeted tax applied to specified products or activities. Depending on the rule, it may be paid upstream by a producer or seller and embedded in the price before the customer sees it. In this lesson's simplified scenario, a designated commercial appliance carries the supplied excise-tax rate.

> 💡 **Strategic Impact**
> Because excise tax raises the cost of acquiring inventory, it directly compresses gross margin. Managers must evaluate **demand elasticity** — whether the market will absorb a price increase — before passing the cost to customers.

When Both Taxes Apply — Never Compound Them

When a lesson scenario states that sales tax and excise tax share one base, calculate each **independently from the original base cost**. Do not apply the second rate to an already-taxed amount.

Worked Example — Meridian Advisory Group

Meridian Advisory Group purchases a commercial appliance for $6,200. Sales tax: 6.25%. Excise tax: 8%.

Tax Type	Calculation	Amount
Base Cost	Given	$6,200.00
Sales Tax (6.25%)	$6,200 × 0.0625	$387.50
Excise Tax (8%)	$6,200 × 0.08	$496.00
Total Cost	$6,200 + $387.50 + $496	**$7,083.50**

> ✅ **Shortcut**
> Add the two rates first (6.25% + 8% = 14.25%), then multiply: $6,200 × 1.1425 = **$7,083.50**. This works because both rates apply to the same base — they can be combined before multiplying.

Part 3 — Property Tax

What Is Property Tax?

Property tax is collected by local municipalities to fund community services: schools, roads, fire and police protection, public libraries, and infrastructure. It applies to both **real property** (land and buildings) and **business personal property** (movable assets such as computers, furniture, and machinery).

Property-assessment systems vary. In this lesson's training municipality, an **assessment rate** converts market value into **assessed value**, which becomes the taxable base.

> ⚠️ **Triple Net Lease (NNN) Warning**
> In a typical NNN lease, the **tenant** pays property taxes, building insurance, and maintenance in addition to base rent. This can make occupancy cost variable, so managers should review both the lease terms and the municipality's tax outlook before signing.

The Three-Step Calculation

Step	What You Calculate	Formula
1	Assessed Value	`Assessment Rate × Market Value`
2	Tax Rate	`ROUND(Town Budget / Total Assessed Value, 4)`
3	Annual Tax Due	`Tax Rate × Assessed Value`

Worked Example — Meridian Advisory Group

Given: Market Value $820,000 · Assessment Rate 40% · Town Budget $310,000 · Total Assessed Value in Town $4,750,000

Step	Calculation	Result
1 — Assessed Value	0.40 × $820,000	$328,000.00
2 — Tax Rate (raw)	$310,000 / $4,750,000	0.0652631…
2 — Tax Rate (rounded)	`=ROUND(rate, 4)`	**0.0653**
3 — Annual Tax Due	0.0653 × $328,000	**$21,418.40**

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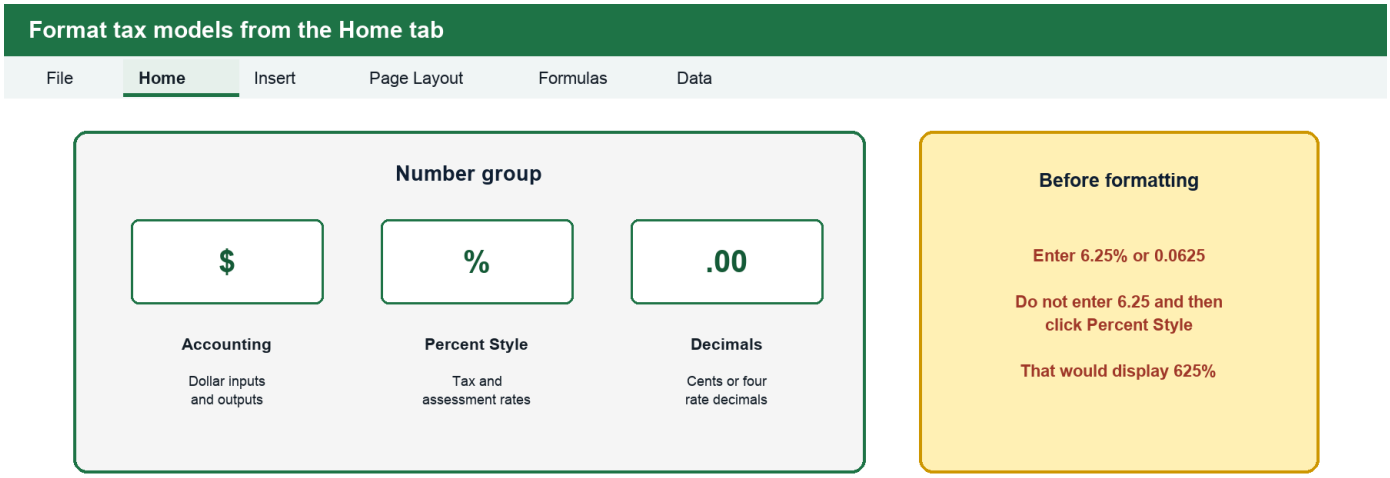
3 · BUILD THE TAX MODELS IN EXCEL

Excel should mirror the order of the business calculation. Keep every assumption in its own labeled input cell, then build formulas that reference those cells. In the examples below, **yellow cells are inputs** and **green cells are formulas**.

> 💻 **Windows and Mac Excel**
> The images show Windows Excel. Mac Excel uses the same commands and formula syntax, although a button may appear in a slightly different position on the ribbon.

Format Dollar Amounts and Rates

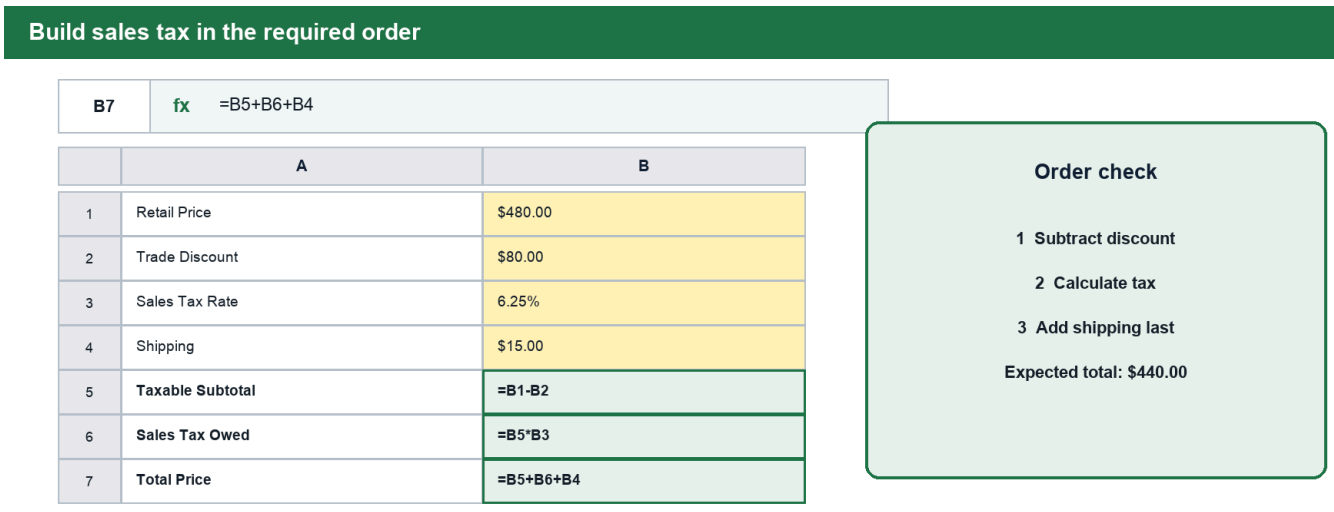
Use **Home** → **Number** to format prices, taxes, and totals as Accounting or Currency. Format tax and assessment rates as percentages. Enter `6.25%` or `0.0625`—do not enter `6.25` and then click Percent Style, because Excel would display `625%`.



Walkthrough 1 — Build the Sales Tax Formula Chain

Set up this model in a blank area of Excel. The cell addresses below are an example; the important requirement is that your formulas reference the cells containing the inputs.

1. **Set up:** In column A, enter the labels Retail Price, Trade Discount, Sales Tax Rate, Shipping, Taxable Subtotal, Sales Tax Owed, and Total Price. In cells B1:B4, enter `$480`, `$80`, `6.25%`, and `$15`.
2. **Select and type:** Select B5 and type `=B1-B2`. Select B6 and type `=B5*B3`. Select B7 and type `=B5+B6+B4`.
3. **Format:** Format B1:B2 and B4:B7 as Accounting or Currency. Format B3 as Percentage.
4. **Check:** B5 should equal **$400.00**, B6 should equal **$25.00**, and B7 should equal **$440.00**. The total must be greater than the taxable subtotal because tax and shipping were added.
5. **Try:** Change the trade discount in B2 from `$80` to `$100`. The taxable subtotal, tax, and total should all update automatically.

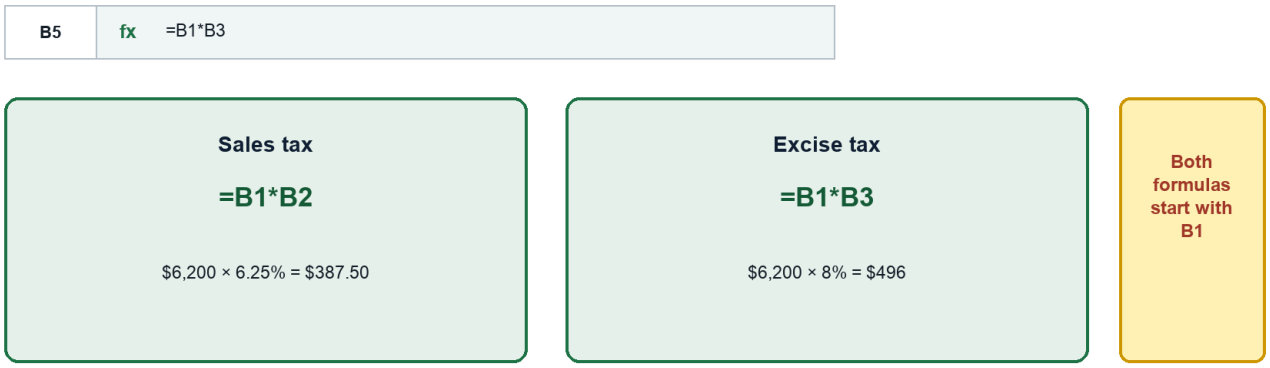


> ⚠️ **Order Matters**
> Subtract the trade discount first, calculate tax second, and add shipping last. Do not calculate tax from the original retail price when a trade discount reduces the taxable base.

Walkthrough 2 — Keep Sales and Excise Taxes on the Same Base

1. **Set up:** Enter Base Cost in A1, Sales Tax Rate in A2, and Excise Tax Rate in A3. Enter `$6,200`, `6.25%`, and `8%` in B1:B3.
2. **Select and type:** In B4, calculate sales tax with `=B1*B2`. In B5, calculate excise tax with `=B1*B3`. In B6, calculate total cost with `=B1+B4+B5`.
3. **Check:** B4 should equal **$387.50**, B5 should equal **$496.00**, and B6 should equal **$7,083.50**.
4. **Confirm with a second method:** In another cell, type `=B1*(1+B2+B3)`. It should also return **$7,083.50**.
5. **Try:** Change the excise tax rate to `9%`. Both the excise tax and total cost should rise, but the sales tax should remain unchanged.

Keep sales and excise taxes tied to the same base cost



If the excise formula begins with the sales-tax-inclusive total instead of B1, the taxes are being compounded incorrectly. Both tax formulas must point back to the original base cost.

Walkthrough 3 — Round the Property Tax Rate First

1. **Set up:** Enter Market Value, Assessment Rate, Town Budget, and Total Assessed Value in A1:A4. Enter `$820,000`, `40%`, `$310,000`, and `$4,750,000` in B1:B4.
2. **Select and type:** In B5, type `=B1*B2` for assessed value. In B6, type `=ROUND(B3/B4,4)` for the tax rate. In B7, type `=B5*B6` for annual tax due.
3. **Format:** Format B1, B3:B5, and B7 as Accounting or Currency. Format B2 as Percentage. Display B6 as a number with four decimal places so you can verify the required rounding.
4. **Check:** B5 should equal **$328,000.00**, B6 should equal **0.0653**, and B7 should equal **$21,418.40**.
5. **Try:** Calculate a monthly NNN planning reserve in B8 with `=B7/12`. The result should be **$1,784.87** when displayed to two decimal places.

Round the property-tax rate before calculating tax due

B6 fx =ROUND(B3/B4,4)

	A	B
1	Market Value	$820,000
2	Assessment Rate	40%
3	Town Budget	$310,000
4	Total Assessed Value	$4,750,000
5	**Assessed Value**	**=B1*B2**
6	**Rounded Tax Rate**	**=ROUND(B3/B4,4)**
7	**Annual Tax Due**	**=B5*B6**

Use the rounded rate

B6 stores 0.0653

B7 uses =B5*B6

Expected annual tax: $21,418.40

Optional monthly reserve: =B7/12

> ✅ **Workbook Connection**
> In class, use the same modeling habits with **new numbers** on the **Live You Try It** tab of the starter workbook. Attempt each row before checking the **FormulaReferenceCard** tab. Then begin the graded **Class Challenge** tax compliance audit: diagnose each supplied filing, repair the liability formula, calculate the adjustment, and reconcile the corrected total to the control total.

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4 · FORMULA REFERENCE

Formula Name	Expression
Sales Tax — Taxable Subtotal	`= Price − Trade Discount`
Sales Tax — Tax Owed	`= Taxable × Tax Rate`
Sales Tax — Total Price	`= Taxable + Tax + Shipping`
Back-Calculate Actual Sales	`= Total Receipts / (1 + Tax Rate)`
Excise — Each tax separately	`= Base × SalesRate` and `= Base × ExciseRate`
Excise — Combined shortcut	`= Base × (1 + SalesRate + ExciseRate)`
Property — Assessed Value	`= Assessment Rate × Market Value`
Property — Tax Rate	`= ROUND(Town Budget / Total Assessed Value, 4)`
Property — Annual Tax Due	`= Tax Rate × Assessed Value`

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5 · CHECK YOUR UNDERSTANDING

Answer all seven questions before class. Answers appear at the end of this document.

1. [Calculation] Tidal Goods Co. sells an order priced at $600. The buyer receives a $100 trade discount. Sales tax is 6.25%. Shipping is $18. What is the taxable subtotal and the total price?

2. [Concept] A merchant collects $4,200 in sales tax over a week. Is that $4,200 the merchant's revenue? Explain why or why not.

3. [Calculation] Register receipts total $78,000 for the day, inclusive of 6.25% sales tax. What were the actual pre-tax sales?

4. [Calculation] Meridian Advisory Group buys office equipment for $8,500. The training jurisdiction states that sales tax is 6.25%, excise tax is 9%, and both share the original base. Calculate the total cost using both methods.

5. [Concept] Under the supplied same-base rule, why is it incorrect to apply excise tax to the sales-tax-inclusive amount?

6. [Calculation] In the training municipality, a property has a market value of $450,000. Assessment rate: 35%. Town budget: $198,000. Total assessed value in town: $2,600,000. Calculate assessed value, tax rate (to 4 decimal places), and annual tax due.

7. [Strategy] Meridian Advisory Group is considering a Triple Net (NNN) lease for new office space. Why should managers review the municipality's tax outlook and the lease terms before signing? What financial risk can an NNN lease create?

Answer Key · Check Your Understanding

#	Answer
1	Taxable = $600 − $100 = $500. Tax = $500 × 0.0625 = $31.25. Total = $500 + $31.25 + $18 = **$549.25**
2	No. Collected sales tax is a **liability held in trust** for the state government, not earned revenue. Recording it as revenue overstates income.
3	$78,000 / 1.0625 = **$73,411.76**. Collected tax = $78,000 − $73,411.76 = $4,588.24.
4	Method 1: ST = $531.25; ET = $765.00. Total = **$9,796.25**. Method 2: $8,500 × 1.1525 = **$9,796.25**. Both match.
5	Under the supplied same-base rule, both taxes use the **original base price**. Applying the second rate to an already-taxed amount violates the scenario rule and overstates total cost.
6	Assessed Value = 0.35 × $450,000 = $157,500. Tax Rate = ROUND($198,000/$2,600,000, 4) = **0.0762**. Annual Tax = 0.0762 × $157,500 = **$12,001.50**.
7	Under a typical NNN structure, the tenant pays property tax, insurance, and maintenance in addition to base rent. Tax changes can therefore turn occupancy cost into **variable overhead**; the exact exposure depends on the lease terms.

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6 · KEY VOCABULARY

Term	Definition
Sales Tax	A state-imposed consumption tax on the retail sale of goods and services; collected by merchants on behalf of the government.
Trade Discount	A reduction from the retail price given to a business buyer; subtracted **before** sales tax is applied.
Excise Tax	A targeted tax imposed on specified products or activities; the liable party and tax base depend on the governing rule.
Assessed Value	The value used as the property-tax base; in this lesson's training municipality: `Assessment Rate × Market Value`.
Assessment Rate	The percentage of a property's market value that a municipality designates as taxable.
Tax Rate (Property)	The rate applied to assessed value to calculate tax due: `Town Budget ÷ Total Assessed Value`; rounded to four decimal places.
Nexus	The legal connection (physical presence or economic activity) that requires a business to collect and remit sales tax in a given jurisdiction.
Triple Net Lease (NNN)	A lease structure in which the tenant typically pays property taxes, insurance, and maintenance in addition to base rent; exact obligations depend on the lease.
Business Personal Property	Movable, tangible operating assets (computers, furniture, machinery) subject to local property tax.
Demand Elasticity	The sensitivity of customer demand to price changes; relevant to excise tax when businesses must raise prices to maintain margin.